Lecture 4

Model Evaluation and Selection

Outline

- Empirical Error and Overfitting
- Bias and Variance
- Evaluation Metrics
- Validation Methods and Model Selection
- Further Reading

Components of Learning

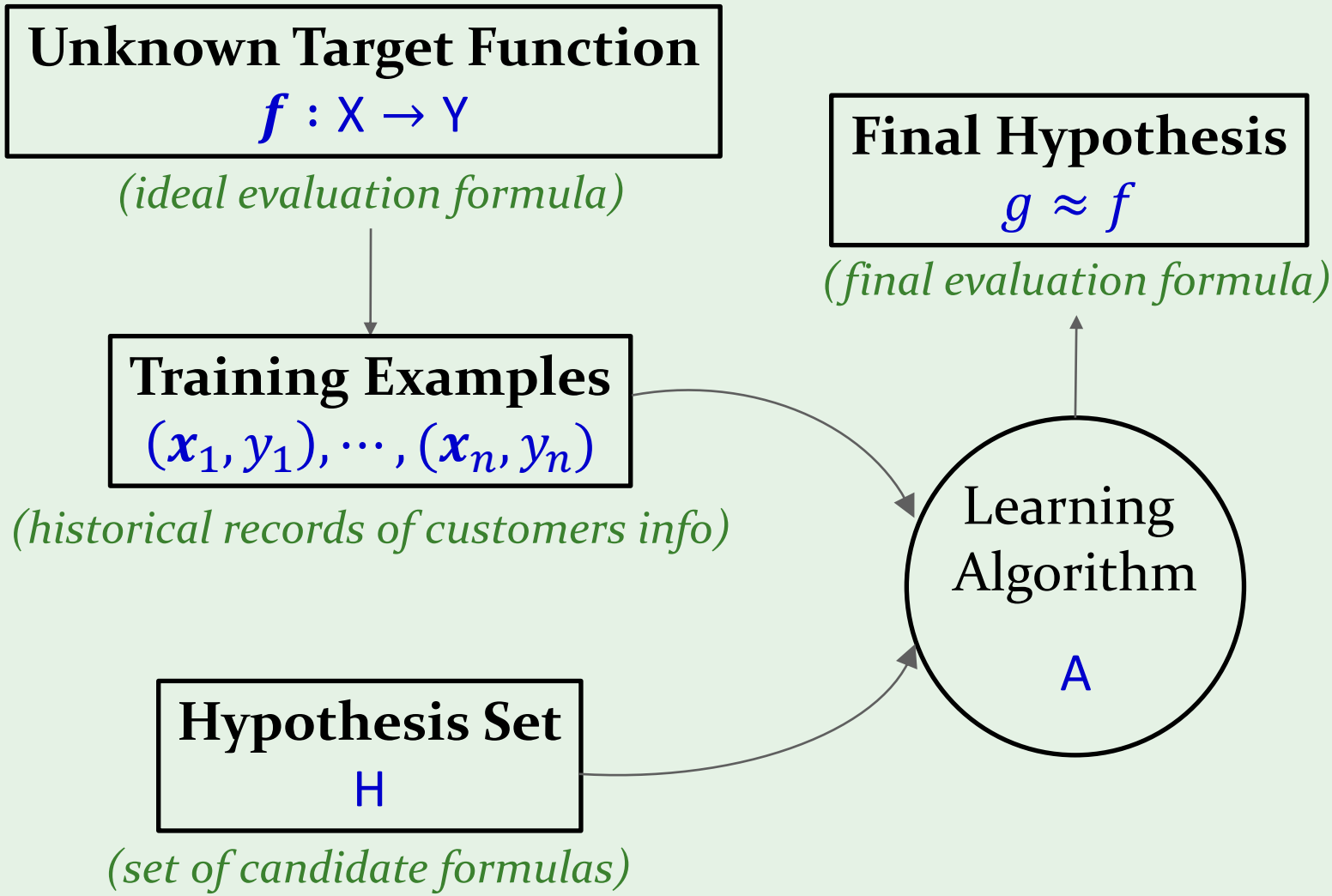
Formalization

- **Input:** $\boldsymbol{x}$ *(customer info)*
- **Output:** y *(good/bad customer?)*
- **Target function:** $f : \mathcal{X} \to \mathcal{Y}$ *(ideal evaluation formula)*
- **Data:** $(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \cdots, (\boldsymbol{x}_n, y_n)$ *(historical records)*



- **Hypothesis:** $g : \mathcal{X} \to \mathcal{Y}$ *(formula to be used)*

Components of Learning



Training, Validation and Test Sets

- **Training set:** learn the weights and intercept for each candidate.
- **Validation set:** compare candidates and choose the regularization strength and threshold.
- **Test set:** estimate performance after all choices are fixed.

Training and Generalization

Training error: average loss on the examples used to fit the model.

Generalization error: expected loss on new examples from the target population.

Test error: an estimate of generalization error from held-out examples.

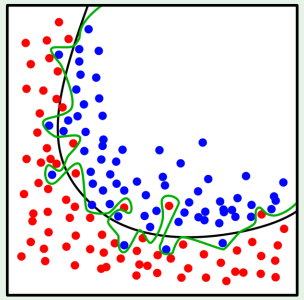
A low training error alone does not tell us how well the model will predict new data.

Empirical Error and Overfitting

- Overfitting:

  When the learner learns the training examples "too well", it is likely that some peculiarities of the training examples are taken as general properties that all potential samples will have, resulting in a reduction in generalization performance.

  - Regularize the training objective
  - Early stop

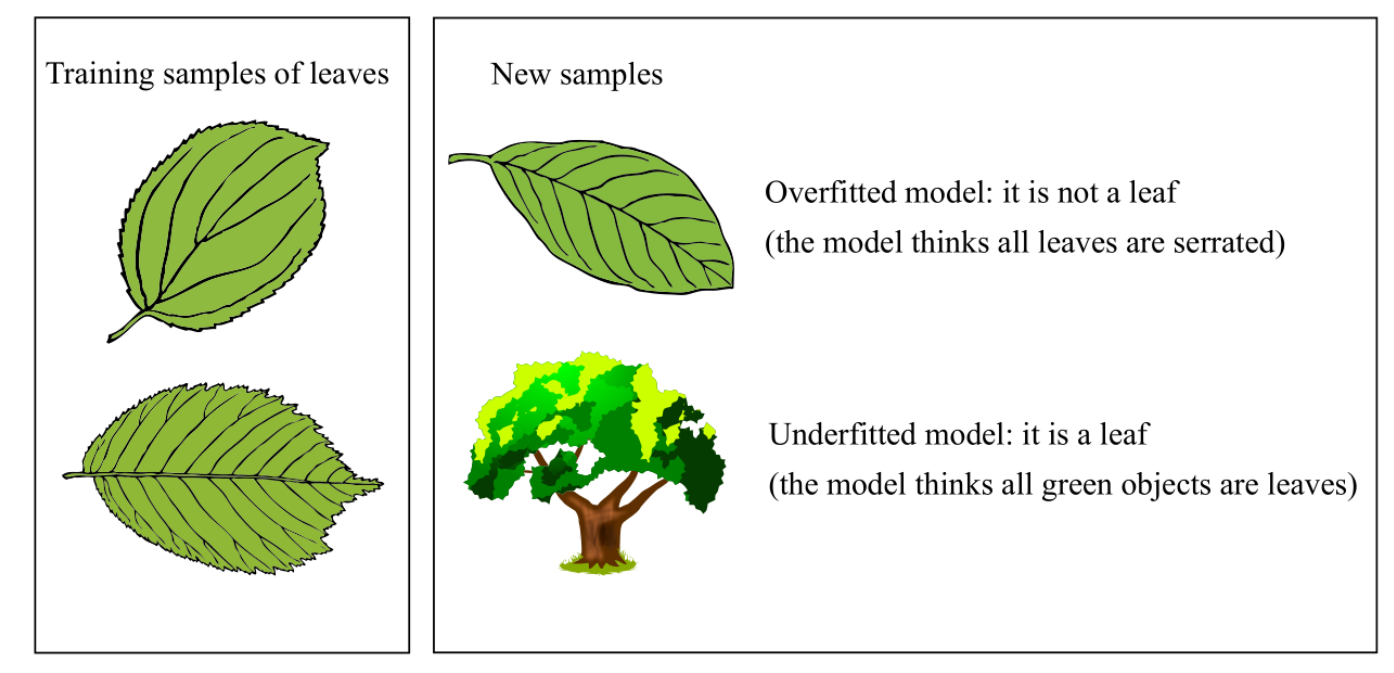


- Underfitting:

  The learner failed to learn the general properties of training examples.

  - Do more branching in decision tree learning
  - Adding more training epochs in neural network learning

Empirical Error and Overfitting



- **Overfitting**: some peculiarities of the training examples are taken as general properties that all potential samples will have.
- **Underfitting**: the learner failed to learn the general properties of training examples.

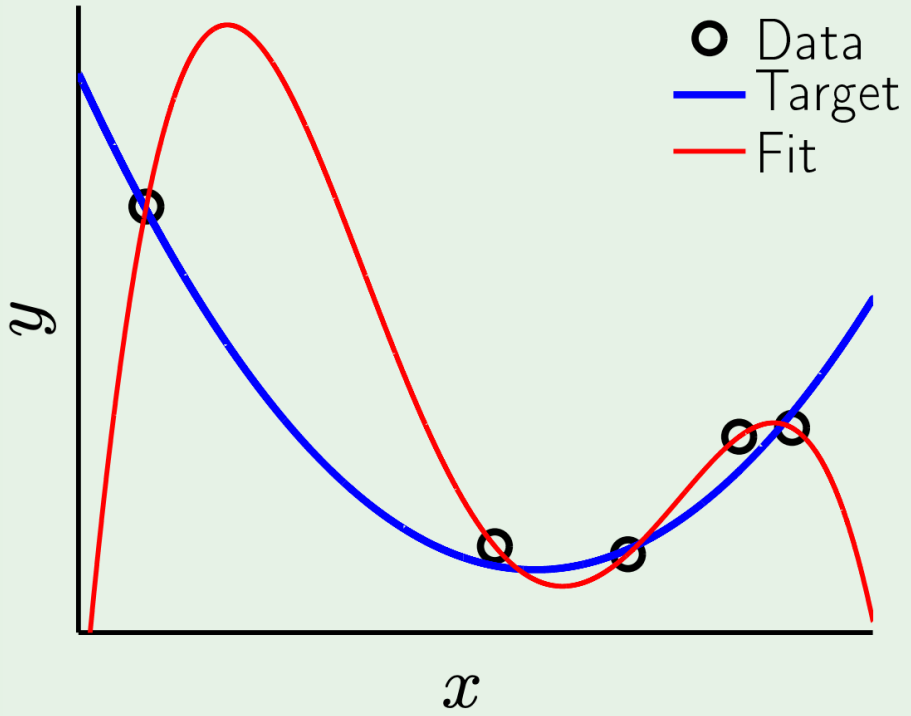
Illustration of over-fitting

Simple target function

5 data points - noisy

4th-order polynomial fit

$E_{\text{in}} = 0$, E_{out} is huge

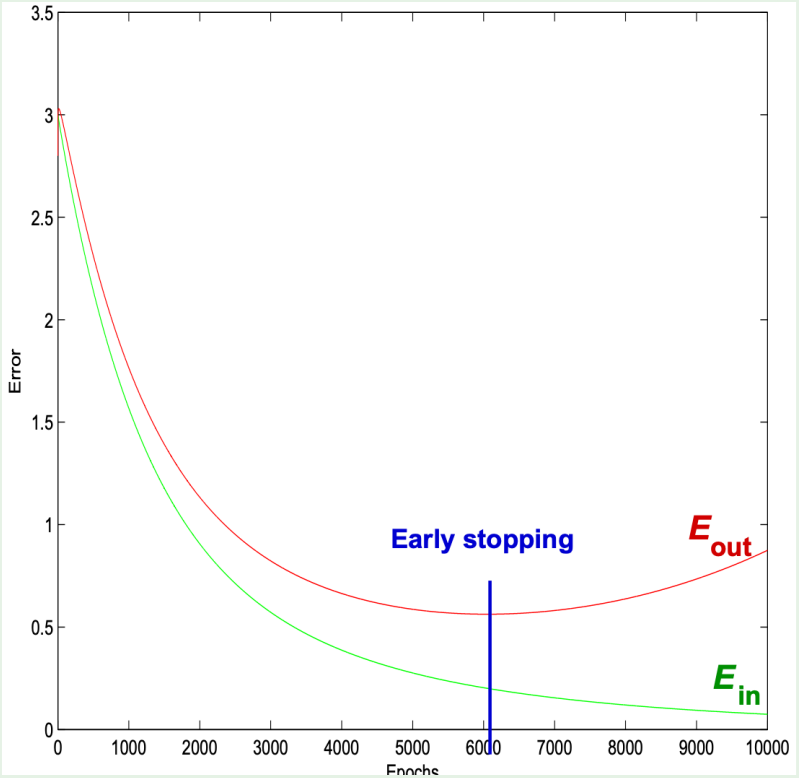


Over-fitting versus bad generalization

Neural network fitting noisy data

Over-fitting: $E_{\text{in}} \downarrow$ $E_{\text{out}} \uparrow$

E_{out} : Generalization Error
or Out-of-Sample-Error



Over-fitting: "fitting the data more than is warranted"

Bias and Variance

- Bias measures the difference between the learning algorithm's expected prediction and the ground-truth label; that is, expressing the fitting ability of the learning algorithm;
- Variance measures the change of learning performance caused by changes to the equal-sized training set, that is, expressing the impact of data disturbance on the learning outcome;

The bias-variance decomposition tells us that the generalization performance is jointly determined by the learning algorithm's ability, data sufficiency, and the inherent difficulty of the learning problem.

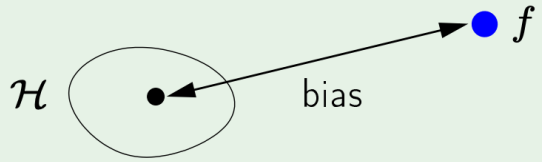
$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

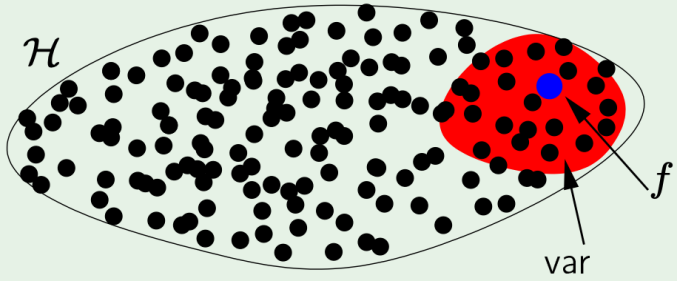
In order to achieve excellent generalization performance, a small bias is needed by adequately fitting the data, and the variance should also be kept small by minimizing the impact of data disturbance.

Bias and Variance Tradeoff

$$\text{bias} = \mathbb{E}_{\mathbf{x}}\left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

$$\text{var} = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2\right]\right]$$

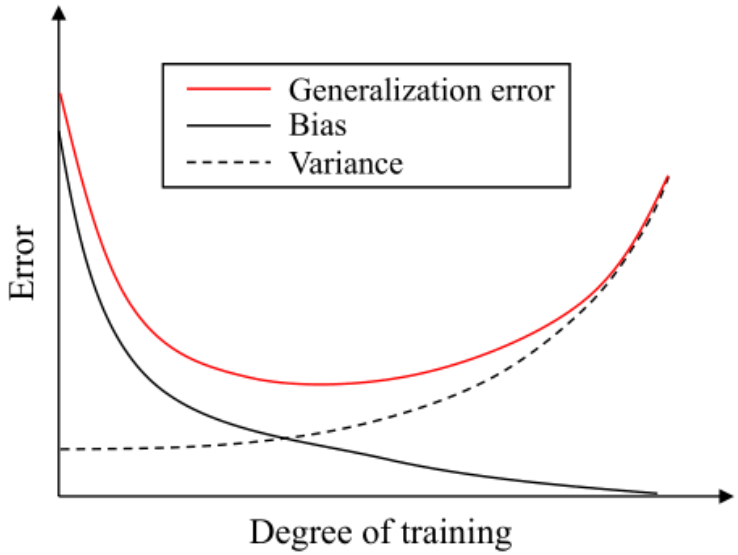




$\downarrow$ $\mathcal{H}\uparrow$ $\uparrow$

Lesson learned

Match the "model complexity" to the data resources, not to the target complexity.



Relationships between generalization error, bias, and variance.

Performance Measure

- Performance measures can quantify the generalization ability. Different performance measures reflect the varied demands of tasks and produce different evaluation results.

- In classification problems, we are given a data set $D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$ where y_i is the ground-truth label of the sample $\boldsymbol{x}_i$. To evaluate the performance of a learner f, we compare its prediction $f(x)$ to the ground-truth label y.

- For regression problems, the most commonly used performance measure is the Mean Squared Error (MSE):

$$E(f; D) = \frac{1}{m} \sum_{i=1}^{m} (f(\boldsymbol{x}_i) - y_i)^2$$

Performance Measure

Error rate and **accuracy** are the most commonly used performance measures in classification problems :

- Error rate is the proportion of misclassified samples to all samples
- Accuracy is the proportion of correctly classified samples instead

Error rate

$$E(f;D)=\frac{1}{m}\sum_{i=1}^{m}\mathbb{I}\left(f\left(\boldsymbol{x}_i\right)\neq y_i\right)$$

Accuracy

$$\begin{aligned}\mathrm{acc}(f;D) &= \frac{1}{m}\sum_{i=1}^{m}\mathbb{I}\left(f\left(\boldsymbol{x}_i\right)=y_i\right)\\ &= 1-E(f;D)\ .\end{aligned}$$

Performance Measure

- We often want to know "*What percentage of the retrieved information is of interest to users?*" and "*How much of the information the user interested in is retrieved?*" in applications like information retrieval and web search. For such questions, **precision** and **recall** are better choices.

- In binary classification, there are four combinations of the ground-truth class and the predicted class, namely *true positive*, *false positive*, *true negative*, and *false negative*. The four combinations can be displayed in a confusion matrix.

The confusion matrix of binary classification

Ground-truth class	Predicted class	
	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Precision $P = \frac{TP}{TP + FP}$

Recall $R = \frac{TP}{TP + FN}$

Six Emails to Classify

- **Illustrative spam filter:** positive means spam. Rank six emails by predicted spam probability. Use threshold 0.75.

Email	Probability	True class	Prediction at 0.75
A	0.90	Positive	Positive
B	0.80	Negative	Positive
C	0.70	Positive	Negative
D	0.60	Positive	Negative
E	0.40	Negative	Negative
F	0.20	Negative	Negative

- True labels stay fixed when the threshold changes.

Confusion Matrix and Accuracy

- At threshold 0.75, A and B are predicted positive.

	Actual positive	Actual negative
Predicted positive	TP = 1 (A)	FP = 1 (B)
Predicted negative	FN = 2 (C, D)	TN = 2 (E, F)

- **True / false:** a correct / incorrect prediction.
- **Positive / negative:** the predicted class.
- Accuracy = (TP + TN) / 6 = 3 / 6 = 0.50.

Precision and Recall

- **Precision:** of the 2 predicted positives, 1 is correct. Precision = 1/2 = 0.50.
- **Recall:** of the 3 actual positives, we found 1. Recall = 1/3.
- Precision checks the spam folder. Recall checks how much of the actual spam we caught.

Quick question: To miss fewer spam messages, should recall increase or decrease?

Precision $P = \frac{TP}{TP + FP}$

Recall $R = \frac{TP}{TP + FN}$

Performance Measure

BEP could be oversimplified, and a more commonly used alternative is **F1-score**:

Harmonic mean (对较小值敏感)

$$F1 = \frac{2 \times P \times R}{P + R} = \frac{2 \times TP}{\text{total number of samples} + TP - TN}$$

The general form of F-score is:

$$F_\beta = \frac{(1 + \beta^2) \times P \times R}{(\beta^2 \times P) + R}$$

$\beta = 1$: Standard F1-score

$\beta > 1$: Recall is more important

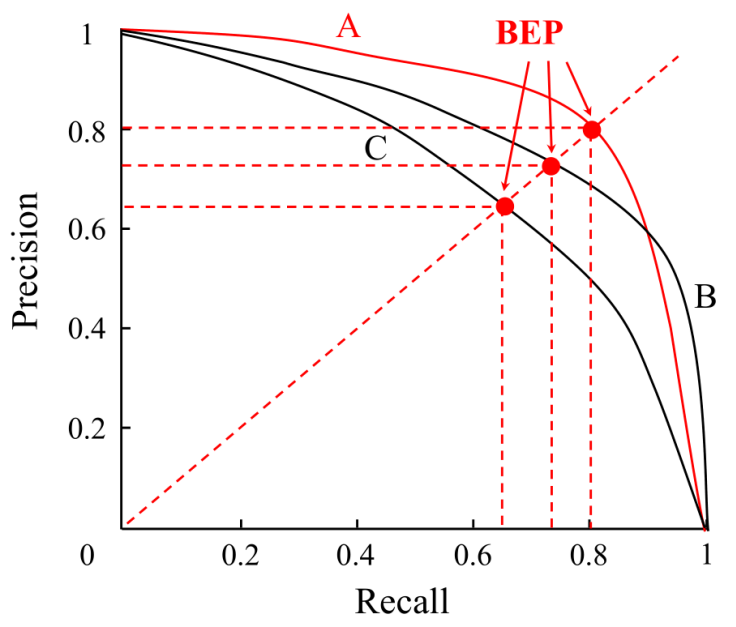
$\beta < 1$: Precision is more important

适用场景举例
垃圾邮件过滤、嫌疑人定罪
一般性模型评估，需要平衡两类错误时
疾病筛查、金融欺诈检测、地震预警

With P = 1/2 and R = 1/3, F1 = 0.40
A high F1 requires both precision and recall to be high.

Performance Measure

We can use the learner's predictions to sort the samples by how likely they are positive. Starting from the top of the ranking list, we can incrementally label the samples as positive to calculate the precision and recall at each increment. Then, plotting the precisions as y-axis and the recalls as x-axis gives the Precision-Recall Curve (P-R curve).



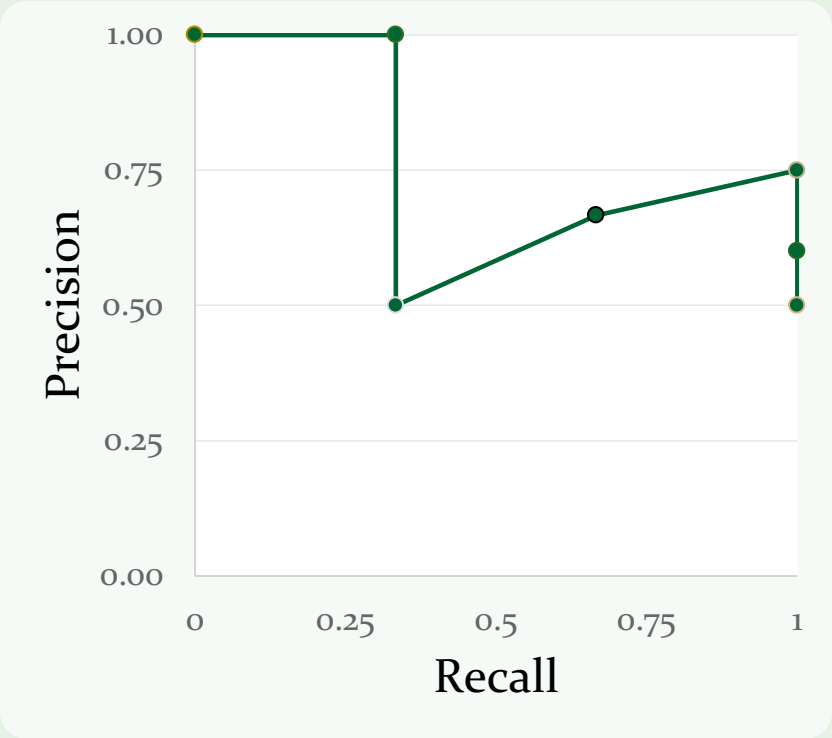
P-R curve and break-even points

- **Break-Even Point (BEP)** is the value when precision and recall are equal, which can be used to compare performance when the P-R curve intersects. （实用性有限）
- Another way of comparing intersected PR curves is to calculate the Area Under PR Curve (AUPR)

Building the Precision–Recall Curve

- Lower the threshold through the ranked scores. Each row gives one point: (recall, precision).

Top k	Recall	Precision
1	1/3	1
2	1/3	1/2
3	2/3	2/3
4	1	3/4
5	1	3/5
6	1	1/2



$$\mathrm{AUPR}_{\mathrm{trap}} = \frac{1}{3}(1) + \frac{1}{3}\frac{1/2 + 2/3}{2} + \frac{1}{3}\frac{2/3 + 3/4}{2} = \frac{55}{72} \approx 0.764.$$

AUPR and Average Precision

AUPR summarizes precision across recall levels（相邻点直线相连）. Average precision (AP) uses step areas（使用后一个点的precision作为矩形的高）:

AP = sum of (increase in recall × precision)

In the six-email ranking, recall increases only at A, C, and D. Each increase is 1/3.

$$AP = (1/3) \times 1 + (1/3) \times (2/3) + (1/3) \times (3/4)$$
$$= 29/36 \approx 0.806$$

AP uses steps. An area computed with straight-line trapezoids can differ.

Performance Measure

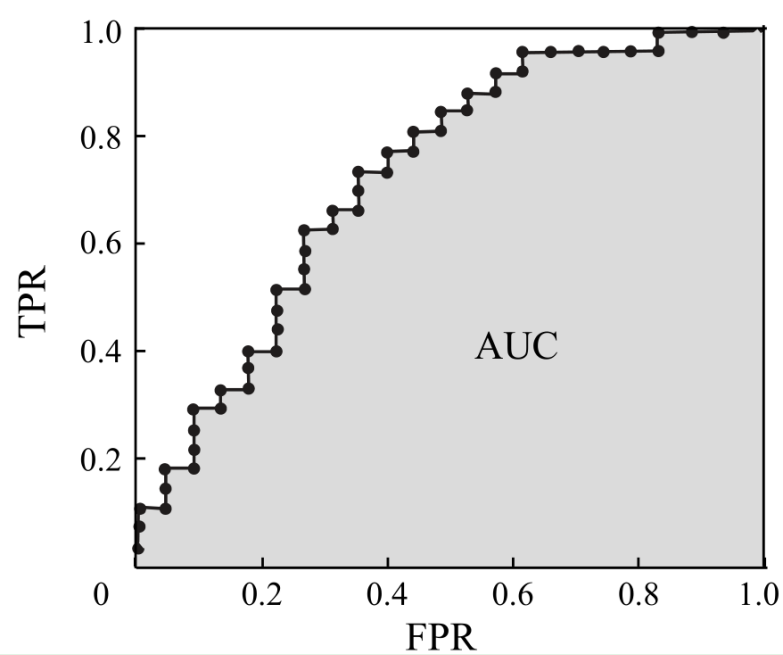
- The ranking quality reflects the learner's "*expected generalization ability*" for different tasks or the generalization ability for "*typical cases*". The *Receiver Operating Characteristics (ROC) curve* follows this idea to measure the generalization ability of learners.

$$TPR = \frac{TP}{TP + FN} = \text{Recall} \qquad FPR = \frac{FP}{FP + TN}$$

- The plotting process is as follows:
  1. given m^+ positive samples and m^- negative samples, we first sort all samples by the learner's predictions, and then set the threshold to maximum, that is, predicting all samples as negative.
  2. At this moment, both TPR and FPR are 0, so we mark at coordinate $(0, 0)$.
  3. Then, we gradually decrease the threshold to the predicted value of each sample along the sorted list, that is, the samples are classified as positive successively. Let (x, y) denote the previous coordinate, we put a mark at $(x, y + \frac{1}{m^+})$ if the current samples is true positive, and we put a mark at $(x + \frac{1}{m^-}, y)$ if the current samples is false positive.
  4. By connecting all adjacent marked points, we have the ROC curve.

Performance Measure

Learner A is better than learner B if A's ROC curve entirely encloses B's ROC curve; However, when there exist intersections, no learner is generally better than the other. One way of comparing intersected ROC curves is to calculate the areas under the ROC curves, that is, Area Under ROC Curve (AUROC or AUC).



ROC curve and AUC with finite samples

Suppose that the ROC curve is obtained by sequentially connecting the points $\{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$, where $x_1 = 0$ and $x_m = 1$.

Then, the AUC is estimated as（梯形面积公式）

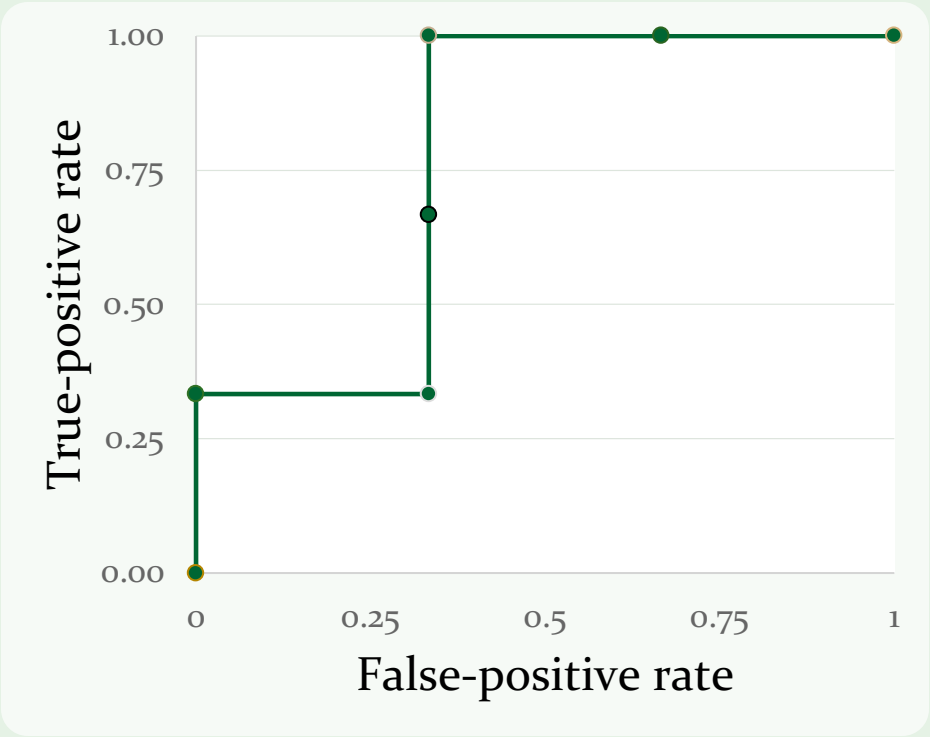
$$\mathrm{AUC} = \frac{1}{2}\sum_{i=1}^{m-1}(x_{i+1} - x_i)\cdot(y_i + y_{i+1})$$

AUC considers the ranking quality of predictions

Building the ROC Curve

- There are 3 positives and 3 negatives. Each added positive moves up by 1/3. Each added negative moves right by 1/3.

Top k	FPR	TPR
0	0	0
1	0	1/3
2	1/3	1/3
3	1/3	2/3
4	1/3	1
5	2/3	1
6	1	1

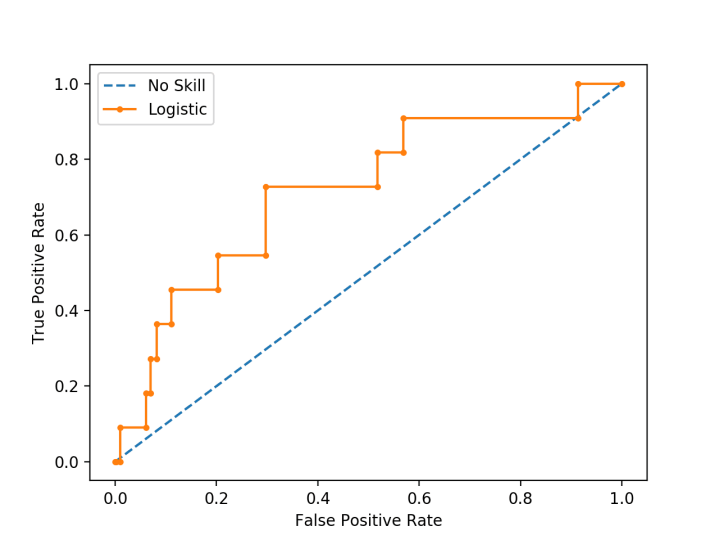


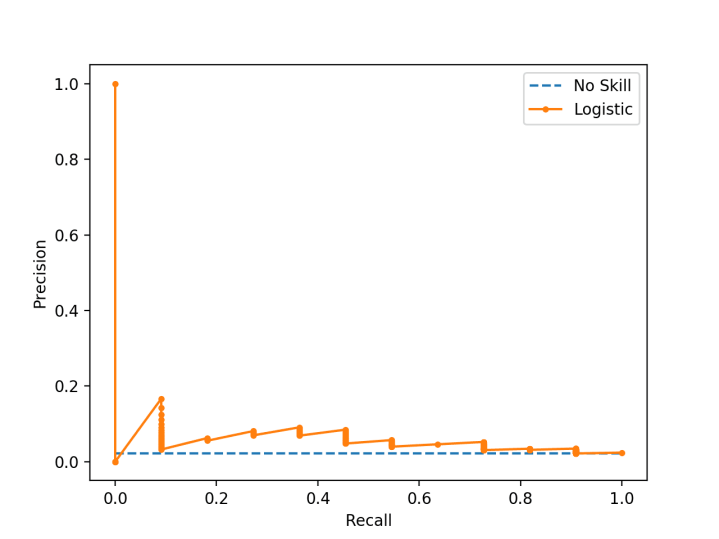
AUROC: Area and Interpretation

- AUROC summarizes how much of the unit square lies below the ROC curve.
- For our staircase, the three horizontal intervals each have width 1/3.
- Their heights are 1/3, 1 and 1. The total area is 1/9 + 1/3 + 1/3 = 7/9.
- A better ranking tends toward the upper-left corner, giving a larger area.

When to Use ROC vs. Precision-Recall Curves?

- ROC curves should be used when there are roughly equal numbers of observations for each class.
- However, ROC is more optimistic when there is a moderate to large class imbalance (less pos., more neg.).





PR makes false alarms visible through precision. It is useful when positive cases are rare.

Cost-Sensitive Error Rate

- In some problems, the consequences of making different errors are not the same, thus we need to assign unequal costs to different errors.

- For binary classification problems, we can leverage domain knowledge to design a cost matrix. where $cost_{ij}$ represents the cost of misclassifying a sample of class i as class j. The larger the difference between the costs is, the larger the difference between $cost_{01}$ and $cost_{10}$ will be.

- With unequal costs, however, we no longer minimize the counts but the total cost, the cost-sensitive error rate is defined as :

$$E(f;D;cost)=\frac{1}{m}\left(\sum_{\boldsymbol{x}_i\in D^+}\mathbb{I}\left(f\left(\boldsymbol{x}_i\right)\neq y_i\right)\times cost_{01}+\sum_{\boldsymbol{x}_i\in D^-}\mathbb{I}\left(f\left(\boldsymbol{x}_i\right)\neq y_i\right)\times cost_{10}\right)$$

		Predicted condition			
	Total population $= P + N$	**Predicted Positive (PP)**	**Predicted Negative (PN)**	Informedness, bookmaker informedness (BM) $= TPR + TNR - 1$	Prevalence threshold (PT) $= \frac{\sqrt{TPR \times FPR} - FPR}{TPR - FPR}$
Actual condition	**Positive (P)**[a]	**True positive (TP),** hit[b]	**False negative (FN),** type II error, miss, underestimation[c]	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{TP}{P} = 1 - FNR$	False negative rate (FNR), miss rate $= \frac{FN}{P} = 1 - TPR$
	Negative (N)[d]	**False positive (FP),** type I error, false alarm, overestimation[e]	**True negative (TN),** correct rejection[f]	False positive rate (FPR), probability of false alarm, fall–out $= \frac{FP}{N} = 1 - TNR$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{TN}{N} = 1 - FPR$
	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{TP}{PP} = 1 - FDR$	False omission rate (FOR) $= \frac{FN}{PN} = 1 - NPV$	Positive likelihood ratio (LR+) $= \frac{TPR}{FPR}$	Negative likelihood ratio (LR−) $= \frac{FNR}{TNR}$
	Accuracy (ACC) $= \frac{TP + TN}{P + N}$	False discovery rate (FDR) $= \frac{FP}{PP} = 1 - PPV$	Negative predictive value (NPV) $= \frac{TN}{PN} = 1 - FOR$	Markedness (MK), deltaP (Δp) $= PPV + NPV - 1$	Diagnostic odds ratio (DOR) $= \frac{LR+}{LR-}$
	Balanced accuracy (BA) $= \frac{TPR + TNR}{2}$	F_1 score $= \frac{2\,PPV \times TPR}{PPV + TPR} = \frac{2\,TP}{2\,TP + FP + FN}$	Fowlkes–Mallows index (FM) $= \sqrt{PPV \times TPR}$	Matthews correlation coefficient (MCC) $= \sqrt{TPR \times TNR \times PPV \times NPV} - \sqrt{FNR \times FPR \times FOR \times FDR}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{TP}{TP + FN + FP}$

Sources: [12][13][14][15][16][17][18][19][20] view · talk · edit

Calibration and ECE

- Calibration asks whether confidence matches the observed fraction of correct predictions.



Conf.	Correct	Acc.
0.60	30/50	0.60
0.90	40/50	0.80

- Each bin has 50 cases. Average the absolute gaps.
- ECE = (0 + 0.10)/2 = 0.05.

Pass@k

Generate k answers per task. Count the task as solved if any answer passes the checks.

Observed pass@k = solved tasks / all evaluated tasks.

Of 100 tasks, the first attempt solves 60; the first 5 attempts solve 80.

Observed pass@1 = 60%. Observed pass@5 = 80%.

This measures whether a passing answer is available. A selection procedure must still choose one answer.

Report k, the checker and the sampling settings.

LLM Agents: Success and Cost

- **Illustrative tool task:** change a booking correctly. Test both agents on the same 100 tasks with a 30-call limit.

Agent	Task success	Mean tool calls
A	85%	20
B	80%	6

- Verify success from the final booking state. Report tool calls, tokens, latency or cost alongside success.
- **Quick question:** Is A better on both measures?
- For RL-trained models, reward, KL and entropy help monitor training. Check task outcomes separately.

Fair Model Comparisons

Use the same validation examples and metric definitions for every candidate model.

Keep training, validation, and final test examples separate. Fit preprocessing using training data only.

Holdout Validation

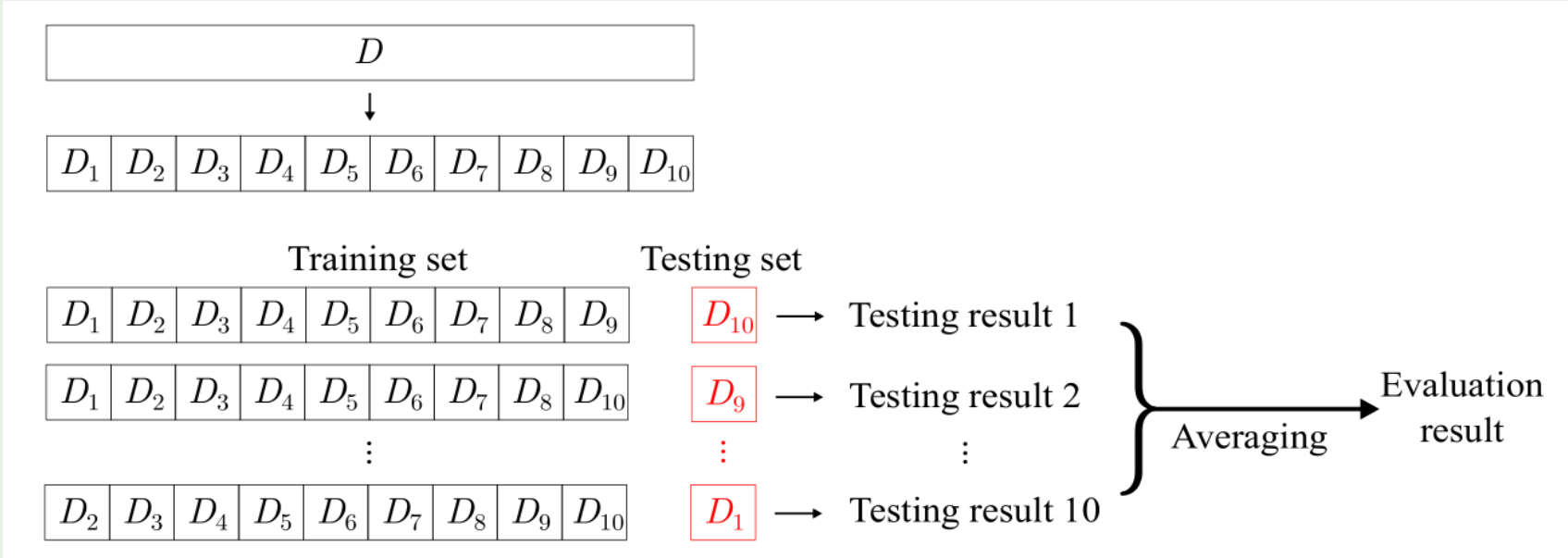
Given the only data set of m samples how can we do both training and testing? The answer is to produce both a training set S and a test set T from the data set D.

- Splits the data set into two disjoint subsets
- The splitting should maintain the original data distribution to avoid introducing additional bias
- We often perform the hold-out testing multiple times, where each trial splits the data randomly, and we use the average error as the final estimation.
- One routine is to use around 2/3 to 4/5 of the examples for training and the rest for testing

Cross-Validation

Cross-validation splits data set D into k disjoint subsets with similar sizes. In each trial of cross-validation, we use the union of $k-1$ subsets as the training set to train a model and then use the remaining subset as the testing set to evaluate the model.

We repeat this process k times and average over k trials to obtain the evaluation result. The most commonly used value of k is 10.



10-fold cross-validation

Leave-One-Out Validation

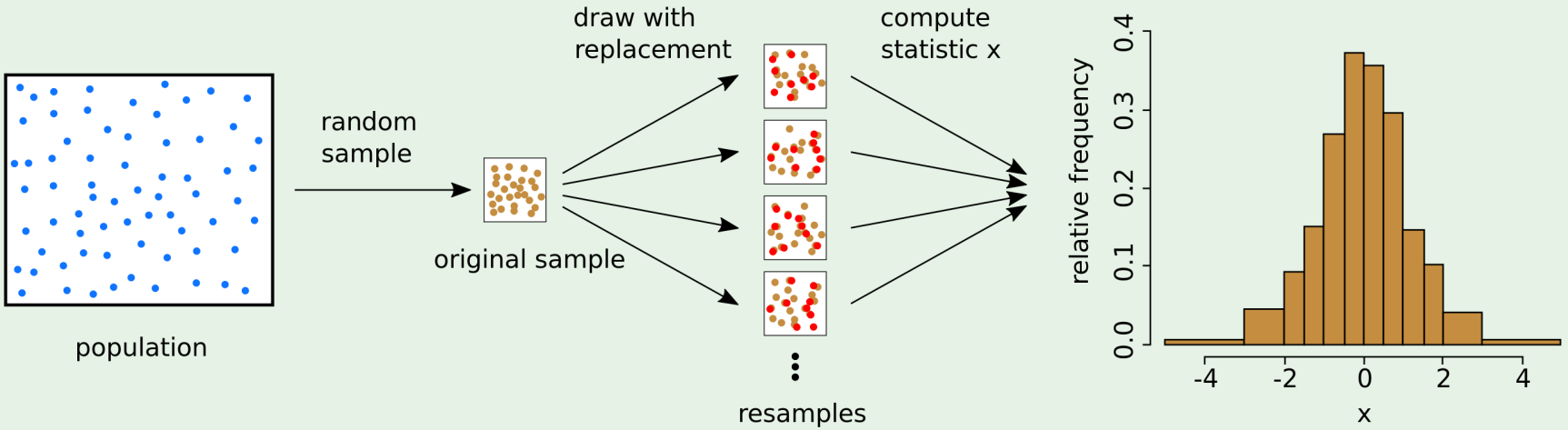
Like hold-out, there are different ways of splitting the data set D into k subsets. To decrease the error introduced by splitting, we often repeat the random splitting p times and average the evaluation results of p times of k-fold cross-validation. For example, a common case is 10-time 10-fold cross-validation.

For a data set D with m samples, a special case of cross-validation is Leave-One-Out (LOO), which lets $k = m$:

- The random splitting does not matter
- The evaluation from LOO is very close to the ideal evaluation
- Computational cost of training m models could be prohibitive for large data sets

Bootstrapping （重采样）

Given a data set D containing m samples, bootstrapping samples a data set D' by randomly picking one sample from D, copying it to D' , and then placing it back to D so that it still has a chance to be picked next time. Repeating this process m times results in the bootstrap sampling data set D' containing m samples. we can use D' as the training set and $D \backslash D'$ as the testing set.



- Roughly 36.8% of the original samples do not appear in the training data.
- Bootstrapping can create multiple data sets, which can be useful for methods such as ensemble learning
- Bootstrapping is particularly useful when the data set is small

Summary

- How to do model selection?
  - Hold-out
  - Cross-validation
  - Leave-one-out
  - Bootstrapping
- How can we evaluate models?
  - Accuracy
  - Precision, Recall, F1
  - AUC、AUPR

Further Reading

- Bootstrap sampling has crucial applications in machine learning, and a detailed discussion can be found in *[Efron and Tibshirani, 1993]*.

- ROC curve was introduced to machine learning in the late 1980s*[Spackman, 1989]*, and AUC started to be widely used in the field of machine learning since the middle 1990s *[Bradley,1997]*. *[Hand and Till,2001]* extended the ROC curve from binary classification problems to multiclass classification problems. *[Fawcett,2006]* surveyed the use of the ROC curve.

- *[Drummond and Holte,2006]* invented the cost curve. Cost-sensitive learning *[Elkan,2001;Zhou and Liu,2006]* is a research topic for learning under unequal cost settings.

Further Reading

- *[Dietterich,1998]* pointed out the risk of using the regular k-fold cross-validation method, and proposed the 5 × 2 cross-validation method. *[Demsar, 2006]* discussed the hypothesis testing methods for comparing multiple algorithms.

- *[Geman et al.,1992]* proposed the bias-variance-covariance decomposition for regression problems, which was later shortened as bias-variance decomposition. For classification problems, however, deriving the bias-variance decomposition is difficult since the 0/1 loss function is discontinuous. There exist many empirical methods for estimating bias and variance *[Kong and Dietterich,1995;Kohavi and Wolpert, 1996; Breiman, 1996; Friedman,1997; Domingos,2000]*.

Bias and Variance

Bias-Variance analysis decomposes E_{out} into

1. How well H can approximate f
2. How well we can zoom in on a good $h \in H$

Applies to **real-valued targets** and uses **squared error**

$$E_{\text{out}}(g^{(\mathcal{D})}) = \mathbb{E}_{\mathbf{x}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

$$\begin{aligned}\mathbb{E}_{\mathcal{D}}\left[E_{\text{out}}(g^{(\mathcal{D})})\right] &= \mathbb{E}_{\mathcal{D}}\left[\mathbb{E}_{\mathbf{x}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]\right] \\ &= \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]\right]\end{aligned}$$

Bias and Variance – Squared Error

Now let us focus on:

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

We define the average hypothesis:

$$\bar{g}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}}\left[g^{(\mathcal{D})}(\mathbf{x})\right]$$

Imagine many data sets $D_1, D_2, \cdots, D_K$

$$\bar{g}(\mathrm{x}) \approx \frac{1}{K}\sum_{k=1}^{K} g^{(\mathcal{D}_k)}(\mathrm{x})$$

Bias and Variance - Using $\bar{g}(\mathbf{x})$

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-f(\mathbf{x})\right)^2\right]=\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})+\bar{g}(\mathbf{x})-f(\mathbf{x})\right)^2\right]$$

$$=\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})\right)^2+\left(\bar{g}(\mathbf{x})-f(\mathbf{x})\right)^2\right.$$

$$\left.+\ 2\ \left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})\right)\ \left(\bar{g}(\mathbf{x})-f(\mathbf{x})\right)\right]$$

$$=\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})\right)^2\right]+\left(\bar{g}(\mathbf{x})-f(\mathbf{x})\right)^2$$

Bias and Variance

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-f(\mathbf{x})\right)^2\right]=\underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})\right)^2\right]}_{\text{var}(\mathbf{x})}+\underbrace{\left(\bar{g}(\mathbf{x})-f(\mathbf{x})\right)^2}_{\text{bias}(\mathbf{x})}$$

Therefore, $\mathbb{E}_{\mathcal{D}}\left[E_{\text{out}}(g^{(\mathcal{D})})\right]=\mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-f(\mathbf{x})\right)^2\right]\right]$

$$=\mathbb{E}_{\mathbf{x}}[\text{bias}(\mathbf{x})+\text{var}(\mathbf{x})]$$

$$=\text{bias}+\text{var}$$